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Hägglund, Susanne; Andtfolk, Malin; Rosenberg, Sara; Wingren, Mattias; Estman, Linda

Published in:
AI and Ethics

DOI:
10.1007/s43681-026-01212-1

Published: 16/07/2026

Document Version
Final published version

Document License
CC BY

Link to publication

Please cite the original version:
Hägglund, S., Andtfolk, M., Rosenberg, S., Wingren, M., & Estman, L. (2026). Applying a care-centered ethical framework when designing robot-assisted medication counselling: a community pharmacy design case study. *AI and Ethics*, *6*, Article 402. https://doi.org/10.1007/s43681-026-01212-1

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AI and Ethics (2026) 6:402
https://doi.org/10.1007/s43681-026-01212-1

RESEARCH

Applying a care-centered ethical framework when designing robot-assisted medication counselling: a community pharmacy design case study

Susanne Hägglund[1] · Malin Andtfolk[2] · Sara Rosenberg[2] · Mattias Wingren[1] · Linda Estman[3]

Received: 3 March 2026 / Accepted: 10 June 2026 / Published online: 16 July 2026
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Abstract

Recent developments in artificial intelligence (AI) have propelled organizations and researchers to call for ethical, responsible, and sustainable deployment of AI-enhanced systems in critical contexts such as healthcare. While legal frameworks like the European Union AI Act are in force, governing AI use and emphasizing ethics, there are a wide range of frameworks and guidelines with no legally binding force that underscore ethics when designing, deploying, and evaluating socially beneficial AI systems. Examples include ethics guidelines for trustworthy artificial intelligence, human-centered AI, and value-sensitive design. AI in community pharmacies is a comparatively nascent field. Nevertheless, research and design initiatives exploring its potential to transform pharmacy practice are subject to the same ethical requirements. Despite the abundance of ethical frameworks, guidance on their operationalization into concrete activities is comparatively scarce. We seek to address this research gap by demonstrating how a value-sensitive design ethical framework can be implemented within research-through-design projects concerning conversational AI on a robot platform. We describe how we employed the care-centered ethical framework while exploring robot-assisted medication counselling in Finnish community pharmacies. We outline phases and activities carried out within the steps of the framework, with a special focus on the pharmacy environment. We conclude by recommendations for applying the framework in processes involving socially assistive robots. The contribution of the paper is an overview of how the Care-Centered framework can be operationalized, alongside an evaluation of its applicability. Additionally, the study examines the relationship between care ethics and socially assistive robots within a real-world healthcare setting.

Keywords Social robotics · Value-driven design · Ethical framework · Community pharmacy · Technology ethics · AI ethics

✉ Susanne Hägglund
susanne.hagglund@abo.fi

Malin Andtfolk
malin.andtfolk@abo.fi

Sara Rosenberg
sara.rosenberg@abo.fi

Mattias Wingren
mattias.wingren@abo.fi

Linda Estman
linda.estman@uis.no

[1] Experience Lab, Institution for social sciences, The Faculty of Human and Social Sciences, Åbo Akademi University, Vaasa, Finland

[2] Department of Health Sciences, The Faculty of Science and Engineering, Åbo Akademi University, Vaasa, Finland

[3] Department of Caring and Ethics, The Faculty of Health Sciences, University of Stavanger, Stavanger, Norway

1 Introduction

At the intersection of recent advances in artificial intelligence (AI) to support healthcare, economic constraints in the pharmacy sector, and staff shortages, attention is turning to the potential of AI-enhanced socially assistive robots (SARs) to co-perform pharmaceutical tasks alongside staff and clients (European Commission, Apply AI Strategy). The deployment of AI systems in community pharmacies has both potential benefits [1] and numerous barriers, including ethical concerns such as securing transparency, upholding integrity and privacy, and low acceptance among clients and staff [2, 3]. Significant ethical challenges arise as AI systems carry out tasks in social settings [4]. It is therefore imperative that ethics is considered, for example, by applying an ethical framework, when assessing the potential of deploying AI systems in these environments [3, 5]. It is requested that actors such as researchers and developers [6], care providers (European Union, [7]), educational bodies [8] and authorities (Committee for the Future, [9]; [10]) mitigate the risks that the deployment of AI systems may cause. Both short and long-term ethical questions must be addressed already at the design stage [21] as well as after implementation [12].

Apart from legislative regulations, there are to date no established design standards or normative guidelines for designing SARs in a pharmacy context, albeit several organizations have launched initiatives to establish general ethical AI principles [4]. A theoretical approach to technology design aligned with human ethical values is Value Sensitive Design (VSD), originating in the field of Human-Computer Interaction (HCI).

We applied the Care-Centered Framework [21], that builds upon VSD, in the early design process of a SAR, as ethics is fundamental to pharmacy practices, as expressed in codes of ethical conduct (Farmasialiitto [13]). Our aim was to explore how a care-centered ethical framework may be translated into concrete activities and to carry out said activities in ethical evaluation of a Research-through-Design (RtD) robot application.

This paper describes the iterative, prospective ethical evaluation of a robot to provide potential clients with medication counselling in a Finnish community pharmacy. Our ambition is to allow for ethically grounded design choices by prospectively applying the Care-Centered (CC) framework throughout the design process. Research shows that methods for translating abstract ethical frameworks into concrete practice do not exist in large numbers [14–17]. In this paper, we aim to reduce that gap by illustrating how a value-sensitive design ethical framework may be applied in RtD projects in healthcare. To the best of our knowledge, no other article has applied and operationalized the CC framework to robot-assisted medication counselling in a community pharmacy setting. The primary contribution of this paper is to provide an overview of how the CC framework can be operationalized in a pharmacy setting and to evaluate its applicability within a RtD project. It further contributes by examining the relationship between care ethics and SARs in a real-world healthcare context.

The paper begins by outlining the Care-Centered Value-Sensitive Design (CCVSD) methodology after which we present the steps of the CC framework, and the activities that were carried out. Next, we present the findings of the ethical evaluation concerning the context in more detail to provide the reader with a thorough example of the value analysis. Lastly, we conclude by discussing lessons learned and providing recommendations.

2 Background

The point of departure in VSD is that technology embodies values [17, 18] and therefore the approach systematically explores and integrates ethical values in the design of technology [19]. It offers a method that systematically integrates human values in design processes of technology by investigating stakeholders' values and mirroring technological artefacts in them ([20]; Umbrello & van de Poel [46]). The VSD approach has also been applied to identifying stakeholder values in human-robot interaction case studies (e.g., Poulsen et al. [18].

Wynsberghe [21] builds upon VSD when she proposes a framework to be used for including ethics in the design process of care robots. The framework distinguishes itself from the general VSD approach as it particularly highlights the specific care context and the fundamental care values and ethical practices, inherent in care work.

2.1 CCVSD approach

VSD is an approach that has been utilized to co-create technology with stakeholders in many different design processes and by many disciplines. It underpins the CCVSD methodology, proposed by van Wynsberghe [21, 35] to understand, design for, and promote core care values in human-robot interaction. Healthcare is a practice where values such as dignity and the safety of vulnerable patients are of utmost importance. Therefore, ethical reflection and the evaluation of values during the design phase of robots provide immediate and tangible benefits by acting as guard rails to protect fundamental values in a high-stakes context such as healthcare. In the wake of the digital transformation, values are shifting and constantly shaping us and our social practices [4]. On a more systemic level, adopting a CCVSD

perspective allows for a broader discussion on which values we judge to be important and what kind of care we want in the future.

Wynsberghe [21] refers to the work of care ethicist Joan Tronto [22] when defining relevant care values to be designed into care robots. The normative criteria include attentiveness, responsibility, competence, and responsiveness, manifested both as a caring attitude and as caring activities in good care practice. Each of the four criteria may be used to evaluate care practices, and each simultaneously corresponds to four phases of the care practice: caring about, care taking, care giving, and care receiving. Attentiveness denotes a care giver's competence to recognize patients' needs, whereas responsibility refers to ensuring that said need is assessed and tended to. This is a typical task where technology increasingly supports humans. Competence refers to the skills with which care is given, and responsiveness is attributed to the patient as a care receiver in the reciprocal nurse-patient relationship.

To explore these values in care practice, the CC framework was developed, aiming to foster ethical reflection in robot designers, to encourage ethical reflection found in the care ethics tradition, and to highlight the relationship between the technical content of a care robot and the manifestation of care values within a care practice ([21]; [35]). The aim of our work is to explore how the framework may be applied and to illustrate its translation into practical activities in a RtD project.

2.2 CC framework

The CC framework [21, 35] may be used by robot designers in the robot design process as a prospective evaluation tool, as well as retrospectively to assess the impact of the robots on care values. Five components are evaluated in the framework, (1) context, (2) practice, (3) actors involved, (4) type of robot, and (5) manifestation of moral elements, aiming to apply to a broad range of robot use cases. The framework applies a comparative approach as values before and after deployment of robots, or between robots, are observed.

When utilizing the framework, the designer follows an outline of steps, not necessarily in a linear, but oftentimes iterative, order [35]. The six steps include data collection, value analyses, scenario comparisons, and recommendations based on the scenario comparisons (see Table 1).

Table 1 The six steps of the Care-Centered framework as proposed by [35]

Step	Activities
1. Data collection of care practice before deployment of robot	Description of the five components of the care practice and researching relevant literature
2. Value analysis of care practice before deployment of robot	Detailed description of care values manifested in the care practice
3. Data collection of robots	Description of the capabilities and appearance of the robot. All technology readiness levels of the robot apply
4. Value analysis of care practice after deployment of robot	Identical to step 1 but with a robot present in the care practice, for comparison. May be carried out in the field or in a laboratory setting, due to emerging technology or prototype status
5. Scenario comparison	Comparison of manifestations of values and value analysis
6. Recommendations for design, implementation, and/or policy	If carried out in early phases of design processes, the recommendations may inform robot design

These steps are outlined in more detail in Sect. 4 where we illustrate how we applied the CC framework and its steps in the design of robot-assisted medication counselling. First, a brief overview of the robot use case is needed to provide context.

3 Robot-assisted medication counselling

The research project PharmAInteraction is a multidisciplinary effort to explore if and how SARs may assist in medication counselling of emergency contraception (ECPs) at community pharmacies in Finland (PharmAInteraction 24). Researchers representing caring sciences, information technology, cognitive science in Finland and Norway and pharmacies, have adopted a RtD approach [45] where a robot application is co-designed with stakeholders to deepen the understanding of the potential of SARs in a pharmacy context. AI has been deployed in decision-making support, optimization, and predicting stock and medication usage [1], and more recently also in brief interactions to support counselling on a range of medicines to clients [32]. In Finland, ECPs are over-the-counter medications that require additional counselling regarding potential side effects, function, use, and individual needs. During the project, the conversational robot Furhat was assigned the role of a standalone Swedish-speaking ECP counsellor with no pharmacist in the immediate loop (See PharmAInteraction video [23] of the designed use case).

The main goal of the project is not to design and develop a robot application for market deployment, but instead to generate and deepen knowledge through a co-design process with stakeholders. This is in line with the overall perspective of RtD that the project applies [24]. Co-design is also in itself an ethical stance and experience, informing research on whether technology may lead to virtuous practices (Hägglund et al., [28]).

We applied the CC framework in the design process as a prospective reflection and evaluation of values that

are relevant to the context and use case of robot-assisted medication counselling. In this paper, we describe how we operationalized the ethical CC framework and illustrate the main activities we carried out over the course of four years. The research output of some of these activities are reported to some extent elsewhere (please see Annex 1 for a list of published, or in review stage, studies that the framework activities also rely on). These studies do not discuss the CC framework and do not share the aim as this paper, focusing on illustrating practical activities. An ethical evaluation using the moral elements of the CC framework from a CCVSD perspective has not been reported elsewhere and is therefore examined more closely in this paper.

3.1 Ethical considerations

We adhere to the tenets of the World Medical Association's Declaration of Helsinki [11] when applying the framework in the design process and ethical research practices (TENK, [42]). These ethical principles also align with the care-centered values guiding our methodological framework. All participants provided their written informed consent to participate in the studies, and data was pseudonymized. Data protection notices were distributed and discussed before the studies. Participation was voluntary and all study participants were informed of the possibility to withdraw without providing further explanation. The observational study and laboratory studies received an ethical review statement from the University Human Sciences Ethics Committee. The study participants were not real pharmacy clients in actual need of ECPs in the role-playing study, and they were not at risk of physiological health injuries as they role-played a scenario in a laboratory without any medication present.

4 Applying the framework

In this section, we present the six steps in more detail based on the description and summary provided in Table 2. Steps 1–2 are reported in greater detail as this data has not been reported elsewhere. The process of applying this framework and carrying out the steps was non-sequential, organic, and iterative as is often the case in co-designing. As the framework allows for other workflows than linear progression, this was not challenging for the ethical evaluation within the team. To facilitate interpretation, Table 2 is structured so that each row represents main activities that were carried out in the six steps. Sample size, methods for data collection and analysis are included where applicable.

4.1 Step 1: data collection of care practice

Owing to the multidisciplinary composition of the research team, creating a depiction of care practice in Step 1 was perceived as straightforward by the robot designers. Knowledge on roles, responsibilities, and task flows in the pharmacy was described and shared in the team. However, to avoid biases and familiarity blindness, activities were carried out on theoretical and empirical levels by non-partisan team members.

On a theoretical level, we conducted a scoping study on available care practices including SARs in medication counselling tasks, which resulted in a zero-sum finding (Andtfolk et al., [25]). We thus concluded that the targeted care practice, community pharmacies in Finland and the Nordics, do not deploy SARs to any degree in the envisioned task. When listing the components of the CC framework, we concluded that the *context* is a community pharmacy, the *practice* revolves around providing counselling on emergency contraception. The *actors* involved include pharmacists and clients, roleplaying either a relative/friend of the client or the client themselves. The *type of robot* is Furhat, a conversational robot using AI models, and the *moral elements* to assess deductively from the CC framework are attentiveness, responsibility, competence, and responsiveness. However, we also used an inductive approach to understand elements expressed by informants in interviews.

At an empirical level, simulations of the care practice were carried out in a research laboratory, where pharmacists interacted with potential clients in an ECP counselling scenario ($N=11$). For ethical reasons, we did not observe pharmacists giving ECP counselling of real clients at a pharmacy but opted for laboratory simulations and role play. The scenarios were recorded and analyzed. The research team divided the task into smaller goals through a Hierarchical Task Analysis (Wingren et al., [31]) to gain a deeper understanding of the care practice, in terms of roles, tasks, interaction patterns between pharmacist and client, and responsibilities. As medication counselling is strictly regulated by law in Finland, this was an important step to consider in our ethical evaluation.

Lastly, we conducted observations of the physical preconditions of pharmacies in the field to complement the picture of the care practice we were researching and designing for, in line with Step 1 of the CC framework. To avoid bias and familiarity blindness, the observations were carried out by other team members than the pharmacist.

Table 2 Main activities carried out in the six steps of the CC framework in the PharmAInteraction research project.

Step	Activities
1.	Description of the care practice through the five components of the CC framework
	Team reflection led by a pharmacist on how the care practice occurs, with the aims and values manifested in the activities
	Data collection of care practice through observations of physical features of three pharmacies in the field
	Scoping review of 2,095 screened studies and ultimately seven studies were included. No previous work on the topic of a conversational robot aiding in medication counselling was available in the scientific literature
	Simulations of the care practice in a research laboratory, where three pharmacists interacted with eight role-playing clients in a care scenario. The Hierarchical Task Analysis method was used to gain understanding of the task of medication counselling, based on recorded video data of the simulations
2.	Deductive (value) analysis based on the four criteria of the CC framework of qualitative interviews with eight role-playing clients and three pharmacists
	Team reflection led by research team members with work experiences of pharmacies and medical counselling. The team shared their knowledge on how the care practice occurs, with the aims and values manifested in the activities
	Deductive analysis of manifested values, based on the four criteria of the CC framework, in three pharmacies, collected by team observation of physical features of pharmacies in the field
	Seven qualitative individual interviews with regulatory authority and sector association representatives, deductively analyzed with the principles of bioethics and inductively analyzed using a reflexive thematic analysis
	Qualitative focus group discussions with eight pharmacists, analyzed via a reflexive thematic analysis
3.	Technical development within the research team has proceeded through numerous stages of development. The first stage resulted in a rule-based robot application. In the second stage, we added a large language model (LLM) with Retrieval-Augmented Generation (RAG) to the application. In the third stage, we constructed an LLM-based multi-agent system (MAS)
	Role playing scenario with potential pharmacy clients and the Furhat robot, in a laboratory setting on two occasions ($n=21$), generating qualitative research data, analyzed using a reflexive thematic analysis and Interpersonal Relationship Model to describe robot appearance and capabilities.
4.	Deductive analysis of manifested values of the CC framework in the pharmacies with the robot use case in mind, collected by team observation of the physical environment of pharmacies in the field.
	Role playing scenario in a laboratory setting with six potential pharmacy clients receiving medication counselling from a robot (two months earlier, they role played the same scenario with pharmacists, see Step 2). Inductively analyzed qualitative interviews using a reflexive thematic analysis.
	Analysis of qualitative interviews with regulatory authority and sector association representatives ($n=7$), exploring comparison of values manifested in the two scenarios (with and without a robot present).
5.	Role playing scenario with potential pharmacy clients and the Furhat robot in a laboratory setting ($n=15$), analyzed using a reflexive thematic analysis and the Qualitative Analysis Guide of Leuven (QUAGOL)
	Analysis of qualitative interviews with pharmacists ($n=8$) where values are compared. Data analyzed via a reflexive thematic analysis.
6.	Team reflection on pathways to sustainable design of SARs, including recommendations on how to design for supporting all three dimensions of sustainability if deploying SARs in pharmacies.
	Analysis of six semi-structured qualitative interviews with reflexive thematic analysis on striking a balance between human autonomy and automation, design recommendations to meet pharmacy clients' needs for human control in robot-assisted medication counseling.
	Joint team reflection sessions, where research results were discussed in terms of design recommendations.

4.2 Step 2: value analysis of care practice before deployment of robot

Currently in Finnish pharmacies, clients who wish to buy ECPs which are non-prescribed over the counter medication, must communicate first with a pharmacist who asks questions and individually evaluates the situation before giving advice, information, and deciding on selling the medicine. Often, the client is taken to the prescription counter for a discussion instead of remaining in the public area of the pharmacy.

Regarding the manifestation of moral elements, listed as criteria in the CC framework, we identified the following in the simulation study with pharmacists and role-playing clients.

Attentiveness (caring about): Acknowledging clients and recognizing their individual needs turned out to be key. This involved seeing and acknowledging the client, intuitively understanding whether they need help and what kind of help, and truly listening to the client in their sensitive role, especially as a relative asking for a friend/loved one. *Responsibility* (caring for): Both informant groups emphasized the importance of non-judgmental, objective, and friendly treatment by staff. Clients expressed strong concerns about privacy, noting that current practices are often insufficient. They also valued care that is clear and transparent, both in terms of the information provided and the physical environment, including opening hours and the organization of aisles.

Competence (care giving): One care value of the pharmacists was to act professionally yet warmly, and to calmly outline potential side effects without scaremongering. In the eyes of a client, this means receiving correct, fact-based information from a knowledgeable and competent pharmacist quickly and swiftly.

Responsiveness (care receiving): Many clients expressed an explicit need for service in their own language, and they appreciated interactivity and the possibility to ask questions.

The findings of qualitative individual interviews with 7 regulatory authority and sector association representatives added to the understanding of moral elements in pharmacies. This study highlights the value of equality to the client service in community pharmacies and legal and ethical standards with respect to both clients and pharmacy personnel, reflecting the fact that the pharmacy sector is highly regulated by law.

A focus group study with eight pharmacists (Rosenberg et al., [29]) further listed medication safety, information quality, and reducing personnel shortage as fundamental values in pharmacies.

A deductive analysis of the physical preconditions of pharmacies in terms of attentiveness, responsibility, competence, and responsiveness was carried out to complete the care practice values. This activity will be illustrated in detail here in Step 2, and later in Step 4.

Two research team members observed three pharmacies using a non-validated observation protocol listing lighting, colors, furnishing and movables, sound environment, and use of windows. They coded the observed factors as either contributing to or hindering the four moral elements. A third research team member representing pharmacy personnel did the same coding exercise as desk research. The activity concluded that most observations were subjectively supporting factors of attentiveness, responsibility, competence, and responsiveness. Table 3 summarizes the observed manifestations of values in the eyes of the three researchers.

The exercise highlights how the pharmacy environment needs to maintain a balance between the privacy (for example, soundproofing) required in client service and the marketing aspects aimed at improving the sales

Table 3 Deductive value analysis of subjective observations of three pharmacies

Value	Total observations (positive/negative)	Examples of codes
Attentiveness	60 (36/24)	+ Windows indicating transparency and openness, neutral color setting, waiting spaces with chairs - Dim lighting, too bright lighting
Responsibility	18 (10/8)	+ Soundproof premises not permitting noise from busy street, low shelves, centrally located, automated medication dispenser use and recycling of medications for client use - Abundance of digital displays, only one cash register for over-the-counter clients
Competence	27 (19/8)	+ Partition walls for integrity and privacy, several cash registers, self-care spaces - Non adaptable chairs and narrow service cubicles, sounds from devices in use
Responsiveness	48 (32/16)	+ Spacious and quiet premises, prescription services allowing for integrity and privacy, spotlights on medication in shelves - Constant low buzzing sound

of other commercial goods (for example, good lighting, digital displays). From a professional ethical standpoint, client privacy is always the focus. The walls between the service counters in the prescription area prevent visibility and dampen sound (often with sound-absorbing materials) and improve the client's privacy. From a client perspective, there is a fine line for example between supportive lighting when browsing pharmacy products and too much lighting hindering readability, and quiet environments allowing for focus and safety on the one hand and jeopardizing integrity and privacy on the other as discussions between staff and clients are overheard.

In general, the observing researchers concluded that the values were manifested as modernity, cleanliness, transparency, honesty, attentiveness to clients' needs, digital curiosity, effectivity, and professionality.

4.3 Step 3: data collection of robot

In Step 3, either the speculative or de facto appearance and capabilities of the robot are described. We focused on the conversational AI robot Furhat in a medical counselling scenario prioritizing high human autonomy—a value highlighted by six potential clients during their interactions with Furhat (Hägglund et al., [30]). The robot stands still throughout the interaction and running the intended application does not require any human input, other than the human interacting with it. Thus, it could be characterized as a human-off-the-loop, in the sense that a human does not have to monitor or manage the agent's counselling when deployed, and no human is involved in the decision process of the robot as to recommending suitable medication. Interviews with potential pharmacy clients testing the application in a laboratory setting however (Hägglund et al., [28]) stressed the need for a "red button" allowing the human interacting with the robot to call for a pharmacist.

In later versions of the robot prototype, employing RAG and an LLM, we added a touchpad next to the robot to assist and facilitate interaction with the robot. The pharmacist also serves as a human in the loop in a sense, as the robot does not sell ECPs, only informs the client of active ingredients, potential side effects, and suggests medication based on client data. The capabilities of the MAS that was designed for the robot were also described (Sabel et al., [32]). Step 3 also includes a comparison between the current robot application that is mainly rule-based with some LLM-integration, and a robot application which uses our LLM-based MAS. This step in the framework also included simulated pharmacy interactions where six Swedish-speaking Finnish women (aged 29–37) engaged individually with the robot Furhat. Using Peplau's [26] Interpersonal Relationship Model as a framework, qualitative interviews were conducted and analyzed using a deductive reflexive thematic anaysis. Peplau's model, describing nursing as a therapeutic, interpersonal process between nurse and patient, was found suitable for studying human-robot interaction, as it emphasizes development of trust, communication, and information as well as patient autonomy.

Furhat is designed as a face with customizable facial animations and replaceable masks that downplay its machine-like appearance. One variable explored in the project is the gaze mediating attention and care. The gaze of a robot is a crucial factor in facilitating successful interactions as it expresses an ethical dimension that is significant for relationships and building rapport (Nyholm et al., [27]) and robot gaze behavior influences clients' trust (Hägglund et al., [28]). Several observations by interviewees were made regarding the potential of Furhat's eyes and gaze to convey care values, particularly attentiveness and responsiveness (Hägglund et al., [28]). We conducted role playing scenarios in a laboratory setting in spring 2024 ($n=15$), with a particular focus on robot gaze. Although this work falls under Step 3, it is carried out late in the project, showcasing the iterative nature of applying the framework. We also collected informants' experiences of robot voice, use of voice, and language, where the analysis is yet unpublished.

4.4 Step 4: value analysis of care practice after deployment of robot

In this step, the same simulations of the care practice as carried out in Step 1 were run in a research laboratory, only this time with a robot, instead of pharmacists, interacting with potential clients in an ECP counselling scenario ($N=6$). We concluded that the components of the framework remained the same, except that the pharmacist was now in the background, available for help and support, and the robot was the main interface interacting with the client. The moral elements were more important when the robot was introduced to the scenario. In the qualitative interviews, the informants enacting this human-robot interaction scenario listed important values such as human autonomy, trust, and trustworthiness of robots, not being judged or shamed, explainability. They also mentioned responsibility when deploying AI systems enacted through a robot, as well as integrity and privacy, patient and medication safety. These values were manifested to various degrees in the robot-assisted medication counselling according to the study participants and five personas were created based on the care value trust, to inform upcoming robot design activities (Hägglund et al., [28]).

Qualitative interviews with seven regulatory authorities and sector association representatives were held, aiming to understand the ethical issues of robot-assisted medication

counselling in their view. A deductive analysis, taking the four principles of bioethics, autonomy, nonmaleficence, beneficence and justice, as a theoretical starting point yielded several values of a care practice where a robot is deployed. A robot might strengthen equity, accessibility of counselling, client autonomy and unburden pharmacists in their daily work, but values such as fair and consistent counselling, medication safety and patient safety may be jeopardized should robots be deployed in medication counselling.

Lastly, reflections were made as to how the physical requirements, observed in Step 2, potentially affected Furhat's deployment and what values were supported or hindered in doing so. A value analysis of an envisioned deployed robot was carried out afterwards by the research team (see Table 4).

Environmental factors, that must be addressed in design and potential deployment, were easily listed, e.g., potentially disturbing sounds from other devices, cashiers, many service points next to each other, lights in marketing screens nearby, information tablets, direct sunlight, mirrors, and spotlights in the over-the-counter medication shelfs. These are variables that might disrupt human-robot interaction. However, their direct effects on care values were not as easily recognized, as we only visualized a robot being present giving medication counselling. We predict, however, that the care value of acknowledging client needs (attentiveness) might be jeopardized if the robot does not function properly due to audio or visual disturbances, for example. Risks of violations of integrity and privacy were more easily noted, in terms of robot placement (responsibility and competence). We also note a risk of discrimination, due to poor accessibility and visibility, for example, of client groups with special needs, challenging the value of competence in giving care.

Envisioning a robot in a pharmacy also involves other impacts on values. A positive effect might involve branding the pharmacy as a modern and professional business due to robot deployment and a visible location in a prime setting of the pharmacy. This is not however easily aligned with the four pre-defined care values. We suggest that attentiveness or responsiveness might be appropriate, as some clients stated in prior interviews that they appreciate and expect a quick, automated check-out cashier (Hägglund et al., [28]). A negative effect might be the location of the robot potentially hindering the personnel's mobility in the pharmacy, possibly affecting the function of providing good care (competence).

4.5 Step 5: scenario comparison

In this step, the researchers compared previous analyses of care practices with and without a robot and assessed impacts on the distribution of roles and responsibilities, the impact on human–human relationships, the generation of new human-robot interactions, and the overall influence on care values.

Table 4 Deductive value analysis of subjective observations of three pharmacies, with an imagined conversational AI robot implemented

Value	Examples of codes
Attentiveness	Robot malfunction due to audio or visual disturbances
Responsibility	Risks of violations of integrity and privacy in terms of robot placement
Competence	Risks of violations of integrity and privacy in terms of robot placement
	Risk of discrimination of client groups with special needs, due to poor accessibility and visibility
	Risk of hindering personnel's unhindered mobility in the pharmacy
Responsiveness	Branding as a modern and professional pharmacy responding to client expectations and wishes

We reflected on these issues in three focus groups with eight pharmacists (Rosenberg et al., [29]) and we found that to them, a pharmacy context supports robot-assisted counselling in attributing quality and trust. Thus, pharmacists and the company behind them (the pharmacy) have an important legitimizing role and responsibility. Robots might alleviate staff shortages currently affecting pharmacy personnel and aid in meetings with aggressive clients, but they might also replace pharmacists, thus diminishing human–human relationships and increasing human-robot interactions if the robot is fully deployed. In terms of care values, the informants argued that pharmacists have skills, for example identifying subtle body language in clients, that the robot lacks for ensuring medication safety. Interestingly, the responsibility of the client to critically evaluate SARs and to know how AI systems work might increase, should robot-assisted medication counselling become a norm. Young clients not being critical enough of the quality of the information provided by a robot was concerning according to the pharmacists of the study.

We are also inductively analyzing data collected in a laboratory setting, where 15 potential pharmacy clients interacted with an updated version of Furhat including the design choices and values expressed in earlier studies. We ran new role-playing scenarios where informants played the role of a potential client, interacted with Furhat and a tablet supporting the dialogue, after which there were interviews and questionnaires using a mixed methods approach.

One activity focuses on the robot's impact on care values, in particular clients' experiences of trust, safety, and security when interacting with the robot. Another activity highlights clients' perceptions of the robot's gaze and whether

the gaze gives rise to indications of caring ethical tension. We are also exploring the liminality of views on what constitutes good care, which is a core care value.

4.6 Step 6: recommendations

All core team members were involved in the activity of translating findings from the empirical data collected across the studies into design decisions. Some contributed more extensively to data analysis and the formulation of design proposals (for example, strengthening customer privacy, introducing a "red button," supporting human autonomy, and the need for multimodal communication via a touchpad), while others were more involved in implementing these design solutions on the robot platform. Nevertheless, all team members participated in the discussions and decision-making processes concerning which features should be iteratively incorporated and tested in subsequent phases.

Additionally, we have shared recommendations mainly for designers and programmers at this point in the process (Hägglund et al., [30]; Wingren et al., [31]; Sabel et al [32]). We also reflected on whether it is possible and justified to design SARs from a sustainability perspective. In a paper on potential pathways to sustainable design of an AI-enhanced, SAR prototype application, we discuss the CC framework as a tool to address economic and social sustainability on a meso and macro level in the design process (Hägglund et al., [33]).

The next steps in this last phase of the CC framework include issuing recommendations and policy briefs from a staff and organizational perspective in relation to SARs in pharmacy settings. The aim is to develop evidence-based guidelines that may guide the implementation and regulation of SARs in pharmacy practice, drawing on speculative design workshops with pharmacists and other pharmacy personnel as well as in-depth interviews with proprietary pharmacists. Survey data has been collected to capture broader perspectives from pharmacy management regarding needs, job demands and resources, and attitudes toward SARs. Furthermore, the robot application integrated with the LLM-based MAS will be further developed and evaluated.

5 Discussion

This discussion will approach the CC framework from four distinct perspectives: the theoretical foundation, methodological considerations, contextual relevance, and the user's perspective while applying it.

5.1 Theoretical foundation

The framework, being the methodology of the care-centered VSD approach, focuses on exploring, identifying, and supporting values in the relationships, roles, and responsibilities of a care practice. It stresses the reciprocal nature of care as the care receiver is seen as an active part of the nurse-patient relationship. Theoretically, the framework is underpinned by care ethics, grounded in feminist scholarship [15], particularly the works of Tronto [35]. In Tronto's work [22], the four stages of the care process, caring about (Attentiveness), care taking (Responsibility), care giving (Competence) and care receiving (Responsiveness), refer to human-human interaction. van Wynsberghe builds upon and extends this in the CC framework by adding non-sentient robots to the practice of care, traditionally considered a strictly human endeavor.

Transposing this model from human-human interaction to human-robot interaction, as is outlined in Steps 1 and 4, runs the risk of attributing true cognition and human characteristics to the AI system. This is context and case dependent and perhaps less likely to be a major issue in a co-bot case, where a team of humans and robots would provide care together, simultaneously. In our design case, where the pharmacy client interacts individually with conversational AI with no human present in the immediate loop, it is hard to assess its *attentiveness*, that is if and how a SAR cares about the client. A socially assistive robot cannot cognitively and emotionally care about humans [34]. The robot may very well succeed in completing a task, all the while being neither intelligent nor aware (see for example [4] who argues that this decoupling of agency and intelligence generates significant ethical challenges such as the values highlighted in our work: i.e. explainability, privacy, responsibility, transparency, and trust). The robot may simulate intelligent behavior that humans may associate with caring and the study participants sometimes refer to the robot "understanding" their language or "seeing" them in a vulnerable situation. Given our ontological position that the robot in question here, Furhat, does not possess cognition or virtue to care about humans, we suggest considering the moral elements in a more passive fashion, as values supported or hindered, instead of actively practiced, by the robot. Unlike traditional care ethics, which assumes human agents only, this paper conceptualizes care as a process where humans and AI systems, embodied in SARs, may interact. However, we argue that the robot is not caring about the human, but it may carry out tasks that support caring practices. According to the CC framework (2013), care taking, or *responsibility* may be delegated to robots to certain degrees, either enhancing care givers' performance or by operating autonomously (e.g., lifting robots) in care practice. Our belief is

that no responsibility should be delegated to the robot but instead to humans designing and implementing the technology [44] and to legislative authorities regulating its use. The robot should be considered a part of a socio-technical context [36]. Therefore, to avoid risking attributing full and moral responsibility to robots, instead of the humans in its loop, we propose careful reflections when carrying out these tasks in the framework.

In our experience however, especially the pharmacists and regulatory informants found it easy to distribute the responsibility to humans and to consider the robot as an instrumental tool working together with humans, unable to bear full responsibility. The power of organizational trust was evident both in interviews with pharmacists and potential clients. They trusted the pharmacy owners, employers, and robot designers to not implement anything that runs the risk of posing threats to medication and patient safety and they trusted that humans had overseen the training of the robot (Hägglund et al., [28]; Rosenberg et al., [29]). We found that sometimes, the robot was perceived as a limited extension of the pharmacy, the pharmacists, and the technical developer, and as a user interface between the care giver and care receiver. Perhaps this contextual embedding in a social system or social institution of care practices is something that an ethical framework could address as a driving factor in value sensitive design. Lastly, we appreciate how the framework treats *responsiveness* as a value and moral element, focusing on active clients and the relational dimension of care values. This applies well to modern Western care practices as many clients are partly more active and informed, expecting a more equal care relationship than before [37], and partly, they are the receiving actors in the care relationship. Their perspective is equally important in a value analysis of care practices.

In value sensitive robot design processes, one should be careful not to model the robot fully on humans describing the moral elements of the care practice and its actors. Instead of fully imitating human traits and interaction designers should acknowledge the otherness of the robot and its strengths that are different from humans' strengths [38]. Task modelling is not affected by this, especially legislatively regulated tasks. In such a scenario, it is meaningful to model robot behavior on tasks carried out by a human in a care practice, as they are governed by law (e.g., right to be informed of prices and alternative options). *Competence*, was easier to assess in the sense that medication counselling is strictly regulated by law in Finland, clearly stating which information must be given to the client.

5.2 Methodology

Methodologically, it is imperative to consider values in human-robot interaction research and design. The idea of technological neutrality has been contested in HCI research [15], in part because of how and with what data systems are trained, and in part because technology deployment shapes humans' lives and values. Furthermore, healthcare and pharmacy domains encompass normative and legal values of good care, rendering ethical evaluations obligatory. Therefore, adhering to ethical frameworks concerns mostly a question of which one(s) to use. We have applied mainly one in our work, the CC framework, and additionally explored value through inductive analysis to enrich the value analysis. Many of our findings are in line with AI ethical frameworks and guidelines, such as the ALTAI guidelines [43], and others.

In terms of operationalizing the methodology, we found Steps 2 and 4 to be closely aligned and simultaneously feeding in to step 5. In the interview studies, we often had a comparative perspective, asking informants to compare care practice with and without a robot and the effect of this on value manifestation. Jumping between the steps is allowed in the CC framework which suits our design case and way of working nicely.

5.3 Context

Contextual factors are important in CCVSD and healthcare as the context manifests values and sets limitations and boundaries for what is possible to deploy. The context in which the care practice is carried out is different across healthcare domains, e.g., emergency units, outpatient units, nursing homes, and pharmacies, which the CC framework addresses throughout the steps. One could question whether a community pharmacy counts as a healthcare context providing care and is therefore eligible for evaluation through the framework. Our starting point has been that it is, although individuals interacting with Finnish pharmacies are primarily referred to as clients or customers [39], suggesting a rather transactional setting. They do, however, acquire medication for their health in pharmacies. The pharmacies also nowadays offer healthcare services, and the personnel provide medication counselling. In line with the fact that "client" is the most used notion for individuals visiting pharmacies [39] and the healthcare providers in Finland (the Wellbeing Services Counties) we choose this notion over the "patient", which is consistently used in the CC framework.

A positive dimension of the CC framework is that researchers may evaluate future robots and their intended features and appearances and visualize deployment of the robot, should testing be impossible in the wild. We

employed laboratory studies and visualizations of the robot in the pharmacy context, instead of in the wild, and we found these activities to be valuable and informative.

5.4 Applying the framework

For researchers and designers applying the CC framework, van Wynsberghe [35] suggests that the framework serves as a starting point for identifying ethical issues relevant to the current design case. This is how we used it as well, as we did not focus only on the four manifestations of moral elements listed by the framework. We also adopted an inductive and bottom-up approach when exploring values and ethical issues relevant to Furhat in a pharmacy setting. We did not perceive this to be at odds with applying the framework, and we agree with its creator that it may serve as a starting point for uncovering ethical issues.

In general, we found applying the CC framework to be beneficial as its theoretical base translates into practical steps quite well. This aligns well with the design and research activities in the project work packages. Instead of being rigid, it allows for flexibility in time (prospective and retrospective ethical evaluations) and between design stages (the steps do not have to be carried out in a chronological order and may be iteratively executed). We chose to assign one team member (first author) to be responsible for the framework adoption, which worked quite well. This person planned and described activities in each step and made sure they were carried out, in cooperation with the project leader.

5.5 Limitations

We acknowledge that our work is contextual and localized and that the applicability of the CC framework is seen through the lens of a Finnish pharmacy setting. While legislation, care setting, and cultural factors will affect the HRI design, thus limiting transferability of the project's findings on how care values may be translated into design, the CC framework is likely to be applicable quite broadly. For example, steps 1–2, aiming to understand the care practice prior to introducing a robot, may be applicable to other healthcare systems and care practices if local laws and norms are followed. The very aim of carrying out the steps is to deepen the understanding of local care practices. Step 4 is likely more challenging, as it is organized around field studies, but the activities may include speculation. Depending on how the framework activities are carried out, methodological limitations may arise due to sample size, and ecological validity may be low if the speculative element is strong. We aim to mitigate these risks through the triangulation of methods and study designs in activities within the CC framework.

6 Conclusion and future work

In this paper, we have illustrated how we applied a care-centered framework for ethical evaluation when designing robot-assisted medication counselling for emergency contraceptive pills. Ethical frameworks for evaluation, with some overlaps, abound and it is challenging for researchers and designers of AI-enhanced welfare technology to identify which one(s) to apply and how to translate them into practical activities. Additionally, the moving target effect of rapidly evolving technology and shifting user expectations adds to the challenge of the relevance and applicability of ethical frameworks to design cases. Our ambition with the paper is to provide insights into how we utilize the Care-Centered Ethical Framework. This is not meant to be a blueprint to copy but rather a guide for other researchers and designers to benchmark, evaluate, and be inspired by in ethical evaluation activities. The main contribution of this paper lies in providing an overview of how the CC framework may be operationalized and an evaluation of its applicability within a RtD project. This study also contributes by discussing the relationship between care ethics and SARs in a real healthcare context.

The importance of AI ethics and acknowledging stakeholders' values when considering the deployment of SARs, is likely to grow and we welcome more research on the application of value sensitive design and ethical frameworks to assist researchers and designers in their work.

Appendix

Studies in the PharmAInteraction project	CC framework steps
A scoping review was conducted of 2.095 screened studies and ultimately seven studies were included. No previous workon the topic of a conversational robot aiding in medication counselling was available in the scientific literature. See Andtfolk et al., [40].	1
Simulations of the care practice were carried out in a research laboratory, where three pharmacists interacted with eight role-playing clients in a care scenario. The Hierarchical Task Analysis method was used to gain understanding of the task of medication counselling, based onrecorded video data of the simulations. See Wingren et al., [31].	1, 6
Seven individual qualitative interviews with regulatory authority and sector association representatives, exploring thereceptiveness of the pharmacy sector regarding the use of social robots for medication counselling purposes in pharmacies. In review.	2, 4

Studies in the PharmAInteraction project	CC framework steps
Seven individual qualitative interviews with regulatory authority and sector association representatives, exploring ethical issues of social robots inpharmacy counselling, analyzed inductively using reflexive thematic analysis. In review.	2
Qualitative focus group discussions with eight pharmacists, analyzed with a reflexive thematic analysis. See Rosenberget al., [29].	2, 5
Implementation overview of a large language model based multi-agent system for medication. See Sabel et al.,[32].	3, 6
Analysis of six qualitative interviews from a Caring Science perspective, concerning potential pharmacy clients'experiences of a robot's gaze. See Nyholm et al., [27].	3
Individual qualitative interviews with six potential pharmacy clients exploring trust in SARs, following a role-playingscenario in a laboratory setting. The data was analyzed using a reflexive thematic analysis. See Hägglund et al., [28].	3, 4
Analysis of six individual, semi-structured interviews utilizing Peplau'sinterpersonal relations model, exploring young adults' needs related to social robots in medication counselling. See Andtfolk et al., [41]	3
Exploring how to strike the balance between human autonomy, human need for control, and robot automation in sixsemi-structured qualitative interviews, analyzed with a reflexive thematic analysis. See Hägglund et al., [30].	3, 6
A collaborative team reflection of possible pathways to sustainable design of an AI-enhanced socially assistive robotprototype application. A call for including sustainability as a tool in the design box and to research sustainable design in social robotics. See Hägglund etal., [33].	6
Fifteen individual qualitative interviews with potential pharmacy clients, analyzed utilizing QUAGOL, exploringhow the robotic gaze in healthcare embodies a caring ethical tension, simulating attentiveness and providing safety while lacking the ethical and relationaldepth of human recognition. In review.	3, 5

Acknowledgements This work was made possible thanks to the dedication and invaluable contributions of the entire PharmAInteraction research team.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by SH, MA, SR, MW, and LE. The first draft of the manuscript was written by SH and all authors commented on versions of the manuscript. All authors read and approved the final manuscript.

Funding Open access funding provided by Åbo Akademi University.

Data availability The datasets presented in this article are not readily available because they contain non-anonymized data that may reveal participant identifiable data. Requests to access the datasets should be directed to the first author.

Declarations

Competing interests The authors declare no competing interests.

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